Week 6: Instruction Sequencing & Control, 8086 Architecture

Midterm Exam Preparation
Week 6

According to your syllabus, Week 6 contains two major parts:

1. Instruction Sequencing & Control
    - Sequential execution
    - Branching concept
    - Instruction sequencing
    - Control signals
    - Hardwired vs. Microprogrammed Control
2. 8086 Architecture
    - Introduction to 8086
    - 16-bit architecture
    - Basic organization
    - Execution Unit (EU)
    - Bus Interface Unit (BIU)
    - Overall processor structure

Let's understand each topic step by step, with examples and important exam questions.

Part 1: Instruction Sequencing & Control

1. What is Instruction Sequencing?

Definition: Instruction sequencing is the process of determining the order in which instructions are fetched and executed by the CPU.

The CPU executes instructions according to a sequence controlled mainly by the Program Counter (PC).

For example, suppose a program contains:

```
Instruction 1: LOAD A
Instruction 2: LOAD B
Instruction 3: ADD A, B
Instruction 4: STORE C
```

The CPU normally executes these instructions in order:

```
1 → 2 → 3 → 4
```

There are two important concepts in instruction sequencing:

- Sequential execution
- Branching

2. Sequential Execution

Definition: Sequential execution is a method in which the CPU executes instructions one after another in their normal order.

The Program Counter (PC) contains the address of the next instruction.

After fetching an instruction, the PC is normally updated to point to the next instruction.

Example

Suppose the following instructions are stored in memory:

Memory Address	Instruction
100	MOV A, 5
101	MOV B, 10
102	ADD A, B
103	STORE C

Execution takes place as follows:

PPT - Understanding Memory Locations and Addresses in Computer Architecture PowerPoint Presentation - ID:2146414
slideserve.com

Step	PC	Instruction
1	100	MOV A, 5
2	101	MOV B, 10
3	102	ADD A, B
4	103	STORE C

The CPU executes each instruction in order.

Important points:

- Instructions execute in normal order.
- PC advances to the next instruction.
- No jump or branch changes the normal sequence.
- It is the simplest form of instruction execution.

3. Branching Concept

Definition: Branching is a process in which the CPU changes the normal sequence of instruction execution by transferring control to another instruction address.

In sequential execution, the PC normally advances to the next instruction. In branching, the PC is updated with a different address.

Why is branching needed?

Branching is used when a program needs to:

- Make decisions.
- Repeat instructions.
- Skip certain instructions.
- Execute different instructions depending on a condition.

Types of Branching

There are two main types:

A. Unconditional Branching

An unconditional branch always transfers control to the specified address, without checking any condition.

Example:

```
100: MOV A, 5
101: JMP 200
102: ADD A, 10
200: SUB A, 2
```

Execution:

1. CPU executes instruction at address 100.
2. CPU reaches address 101.
3. JMP 200 changes the PC to 200.
4. CPU executes the instruction at address 200.
5. Instruction at address 102 is skipped.

Essentials of Microcontroller Use Learning about Peripherals: Programming Part 2 | Renesas
renesas.com

B. Conditional Branching

A conditional branch transfers control to another address only when a specified condition is satisfied.

Example:

```
CMP A, B
JE 200
```

Here:

- CMP compares A and B.
- JE means Jump if Equal.
- If A equals B, the CPU jumps to address 200.
- Otherwise, execution continues with the next instruction.

Other examples of conditional jump instructions include:

Instruction	Meaning
JE / JZ	Jump if Equal / Zero
JNE / JNZ	Jump if Not Equal / Not Zero
JC	Jump if Carry
JNC	Jump if No Carry
JG	Jump if Greater
JL	Jump if Less

These are examples of 8086-style jump instructions, which you will study in more detail later.

Sequential Execution vs Branching

Sequential Execution	Branching
Executes instructions in normal order	Changes the normal execution order
PC normally advances	PC is updated to a branch target
No jump is required	Uses jump or branch instructions
Used for normal program flow	Used for decisions and loops

4. Control Signals

Definition: Control signals are electrical signals generated by the Control Unit to coordinate and control the operations of different CPU components.

They tell the CPU components what operation to perform and when to perform it.

Functions of Control Signals

1. Control data movement between registers.
2. Control memory read and write operations.
3. Select ALU operations.
4. Control instruction execution.
5. Update the Program Counter.
6. Coordinate different CPU components.

Examples of Control Signals

Control Signal	Function
Memory Read	Reads data from memory
Memory Write	Writes data into memory
Register Load	Loads data into a register
ALU Control	Selects an ALU operation
PC Load	Loads a new address into PC
PC Increment	Increases the PC
IR Load	Loads an instruction into IR

Example: Addition Operation

Suppose the instruction is:

```
ADD R1, R2
```

Assume:

```
R1 = 10
R2 = 20
```

The Control Unit generates signals to:

1. Select R1.
2. Select R2.
3. Send both values to the ALU.
4. Select the addition operation.
5. Enable the destination register.
6. Store the result.

```
10 + 20 = 30
```

Remember: Control signals coordinate the CPU components. Without them, the CPU cannot execute instructions in an organized way.

5. Hardwired Control Unit

Definition: A hardwired control unit is a control unit in which control signals are generated using fixed hardware circuits such as logic gates, decoders, and flip-flops.

Its control logic is implemented directly in hardware.

How does it work?

1. The CPU receives an instruction.
2. The instruction is decoded.
3. Fixed logic circuits determine the required operation.
4. The circuits generate the appropriate control signals.
5. The CPU performs the operation.

Introduction to Control Unit and its Design - GeeksforGeeks
geeksforgeeks.org

Advantages

- Very fast operation.
- Control signals are generated directly by hardware.
- Suitable for simpler instruction sets.

Disadvantages

- Difficult to modify.
- Complex to design for complicated instruction sets.
- Changes may require hardware redesign.

Example: A control unit implemented using fixed logic gates and decoders.

6. Microprogrammed Control Unit

Definition: A microprogrammed control unit generates control signals using microinstructions stored in a special control memory.

Instead of generating all control signals through fixed logic circuits, it uses small programs called microprograms.

What is a Microinstruction?

A microinstruction is a small control instruction that specifies the operations required inside the CPU.

A sequence of microinstructions is called a microprogram.

How does it work?

1. The CPU receives an instruction.
2. The Control Unit identifies the required microprogram.
3. Microinstructions are read from control memory.
4. Each microinstruction generates the required control signals.
5. The CPU performs the operation.

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Advantages

- Easier to modify and update.
- Easier to design for complex instructions.
- Flexible control implementation.

Disadvantages

- Generally slower than hardwired control.
- Requires control memory.
- Additional microinstruction processing is needed.

7. Hardwired vs Microprogrammed Control Unit

Important Long Question

Hardwired Control	Microprogrammed Control
Uses fixed hardware circuits	Uses microinstructions
Control signals generated by logic circuits	Control signals generated through microprograms
Generally faster	Generally slower
Difficult to modify	Easier to modify
Complex to redesign	Easier to design for complex instructions
No microprogram control memory required	Requires control memory

Easy way to remember:

- Hardwired = Hardware-based control.
- Microprogrammed = Program-based control.

Exam answer: A hardwired control unit generates control signals through fixed logic circuits, whereas a microprogrammed control unit generates control signals by executing microinstructions stored in control memory. Hardwired control is generally faster, while microprogrammed control is more flexible and easier to modify.

Part 2: Introduction to 8086 Architecture

8. What is 8086 Microprocessor?

Definition: Intel 8086 is a 16-bit microprocessor introduced by Intel in 1978. It is an important early processor in the x86 family.

It is called a 16-bit microprocessor because its ALU and main data-processing registers are 16 bits wide.

Important Features of 8086

Feature	Description
Processor	Intel 8086
Architecture	16-bit
Data bus	16-bit
Address bus	20-bit
Maximum memory	1 MB
Registers	16-bit general-purpose registers
Memory organization	Segmented memory
Instruction queue	6 bytes
Design	BIU and EU

Why does 8086 support 1 MB memory?

8086 has a 20-bit address bus.

The number of possible memory addresses is:

\[2^{20}=1,048,576 \]

Since:

\[1,048,576 \text{ bytes}=1 \text{ MB} \]

Therefore, 8086 can address 1 MB of memory.

Important numerical concept

1 MB

\[2^{20}=1,048,576\text{ bytes} \]

20-bit address bus → 1,048,576 addressable byte locations.

Remember:

- 16-bit processor
- 20-bit address bus
- 16-bit data bus
- 1 MB memory capacity

Do not confuse the 16-bit data bus with the 20-bit address bus.

9. 16-bit Architecture of 8086

Definition: A 16-bit architecture means that the processor's main data-processing operations and general-purpose registers operate on 16-bit data.

A bit is a binary digit: 0 or 1.

A 16-bit register can store:

\[2^{16}=65,536 \]

different bit patterns.

Important characteristics

1. ALU is 16-bit.
2. General-purpose registers are 16-bit.
3. Data bus is 16-bit.
4. Address bus is 20-bit.
5. Memory is organized into segments.

Example

A 16-bit register can store:

```
0000000000001010
```

This represents decimal 10.

A 20-bit address can represent:

```
00000000000000000001
```

The address bus carries memory addresses, while the data bus carries data.

10. Basic Organization of 8086

The 8086 processor is divided into two major units:

1. Bus Interface Unit (BIU)
2. Execution Unit (EU)

8086 microprocessor | PPSX
slideshare.net

These two units work together to execute instructions efficiently.

11. Bus Interface Unit (BIU)

Definition: The Bus Interface Unit (BIU) is the part of the 8086 processor responsible for communicating with memory and input/output devices.

It handles fetching instructions and transferring data between the processor and external devices.

Main functions of BIU

1. Fetches instructions from memory.
2. Reads data from memory.
3. Writes data to memory.
4. Handles input/output operations.
5. Generates physical memory addresses.
6. Maintains a 6-byte instruction queue.

Main components of BIU

Component	Function
Segment Registers	Store segment addresses
Instruction Pointer (IP)	Holds offset of the next instruction
Instruction Queue	Temporarily stores prefetched instructions
Address Generation Circuit	Generates physical addresses
Bus Control Logic	Controls external bus operations

You will study segment registers and address generation in Week 7.

What is the Instruction Queue?

The 8086 has a 6-byte instruction queue.

It temporarily stores instructions fetched from memory so that the Execution Unit can process them.

Why is it needed?

Without an instruction queue, the CPU might have to wait for each instruction to be fetched before executing it.

The BIU can fetch upcoming instructions while the EU executes the current instruction.

Architecture of Intel 8086
chipprogrammer.blogspot.com

This improves processor efficiency.

Important: BIU fetches instructions; EU executes instructions.

12. Execution Unit (EU)

Definition: The Execution Unit (EU) is the part of the 8086 processor responsible for decoding and executing instructions.

It takes instructions from the instruction queue and performs the required operations.

Main functions of EU

1. Takes instructions from the instruction queue.
2. Decodes instructions.
3. Performs arithmetic operations.
4. Performs logical operations.
5. Controls general-purpose registers.
6. Updates status flags.
7. Requests BIU to access memory when required.

Main components of EU

Component	Function
ALU	Performs arithmetic and logical operations
General-Purpose Registers	Store temporary data
Flag Register	Stores status and control flags
Instruction Decoder	Decodes instructions
Control Circuit	Coordinates execution

Intel 8086-8088 Microprocessor & Assembly Language.pptx
slideshare.net

Example: Addition in 8086

Suppose:

```
AX = 10
BX = 20
```

Instruction:

```
ADD AX, BX
```

The following process takes place:

1. EU takes the instruction from the instruction queue.
2. EU decodes the instruction.
3. EU sends AX and BX values to the ALU.
4. ALU performs addition.
5. Result is stored in AX.

```
AX = 10 + 20
AX = 30
```

The EU performs the calculation, while the BIU handles memory and bus communication.

13. Difference Between BIU and EU

Very Important Exam Question

BIU (Bus Interface Unit)	EU (Execution Unit)
Handles communication with memory and I/O	Executes instructions
Fetches instructions	Decodes instructions
Reads and writes data	Performs arithmetic and logical operations
Generates physical addresses	Uses registers and ALU
Contains instruction queue	Contains ALU and instruction decoder
Contains segment registers and IP	Contains general-purpose and flag registers

Easy trick:

- BIU = Fetch and communicate.
- EU = Decode and execute.

14. How BIU and EU Work Together

This is one of the most important concepts in 8086 architecture.

The BIU and EU work simultaneously to improve performance.

Step-by-step working

1. BIU fetches instructions from memory.
2. BIU places fetched instructions into the 6-byte instruction queue.
3. EU takes an instruction from the queue.
4. EU decodes the instruction.
5. EU executes the instruction.
6. Meanwhile, BIU can fetch upcoming instructions if the queue has space.

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slideshare.net

Example

Suppose the program contains:

```
MOV AX, 5
MOV BX, 10
ADD AX, BX
```

The process:

BIU	EU
Fetches MOV AX, 5	Executes previous instruction, if any
Fetches MOV BX, 10	Executes MOV AX, 5
Fetches ADD AX, BX	Executes MOV BX, 10
Fetches upcoming instructions	Executes ADD AX, BX

This overlapping of instruction fetching and execution is called pipelining.

What is Pipelining?

Definition: Pipelining is a technique in which different stages of instruction processing overlap, allowing the processor to work on multiple instructions at different stages simultaneously.

In 8086, the BIU fetches instructions while the EU executes previously fetched instructions.

Benefits:

- Improves processor efficiency.
- Reduces waiting time.
- Increases instruction throughput.

Important: When a branch or jump changes the instruction sequence, the previously prefetched instructions in the queue may become invalid. The queue is flushed, and BIU fetches instructions from the new address.

15. Overall Processor Structure of 8086

The complete 8086 processor consists of two major units working together.

Unit IV 8086 complete ppt, architecture and instruction set.pptx
slideshare.net

BIU includes:

- Segment registers
- Instruction Pointer (IP)
- 6-byte instruction queue
- Address generation circuit
- Bus control logic

EU includes:

- Arithmetic Logic Unit (ALU)
- General-purpose registers
- Flag register
- Instruction decoder
- Control circuitry

Complete working

1. BIU fetches instructions from memory.
2. Instructions are stored in the instruction queue.
3. EU takes instructions from the queue.
4. EU decodes instructions.
5. EU executes arithmetic or logical operations.
6. If memory access is needed, EU requests the BIU to perform it.
7. BIU transfers the required data between memory and processor.

Important Differences for Exams

1. Hardwired vs Microprogrammed Control

Hardwired	Microprogrammed
Fixed logic circuits	Microinstructions
Faster generally	Slower generally
Difficult to modify	Easier to modify
Hardware-based	Control-memory-based

2. Sequential vs Branching

Sequential	Branching
Normal instruction order	Changes instruction order
PC advances normally	PC changes to target address
No jump	Jump or branch instruction

3. BIU vs EU

BIU	EU
Fetches instructions	Executes instructions
Communicates with memory	Performs calculations
Generates physical addresses	Decodes instructions
Has 6-byte queue	Has ALU

Important Long Questions for Midterm

Prepare these questions thoroughly:

1. Explain sequential execution and branching with examples.
2. What is instruction sequencing? Explain its types.
3. What are control signals? Explain their functions.
4. Explain hardwired and microprogrammed control units with diagrams.
5. Differentiate between hardwired and microprogrammed control.
6. What is the Intel 8086 microprocessor? Explain its important features.
7. Explain the 16-bit architecture of 8086.
8. Explain the basic organization of the 8086 processor with a block diagram.
9. Explain the functions and components of the Bus Interface Unit (BIU).
10. Explain the functions and components of the Execution Unit (EU).
11. Differentiate between BIU and EU.
12. Explain how BIU and EU work together in 8086 architecture.
13. What is pipelining? Explain how 8086 uses pipelining.

Important Short Questions

Q1. What is instruction sequencing?

Instruction sequencing is the process of determining the order in which CPU instructions are executed.

Q2. What is sequential execution?

Sequential execution is the execution of instructions one after another in their normal order.

Q3. What is branching?

Branching changes the normal execution sequence by transferring control to another instruction address.

Q4. What is a control signal?

A control signal is an electrical signal generated by the Control Unit to coordinate CPU operations.

Q5. What is hardwired control?

Hardwired control generates control signals using fixed hardware circuits.

Q6. What is microprogrammed control?

Microprogrammed control generates control signals using microinstructions stored in control memory.

Q7. What is 8086?

8086 is a 16-bit microprocessor developed by Intel in 1978.

Q8. What is BIU?

BIU is the Bus Interface Unit, responsible for memory and I/O communication and instruction fetching.

Q9. What is EU?

EU is the Execution Unit, responsible for decoding and executing instructions.

Q10. What is the size of the 8086 instruction queue?

The 8086 instruction queue is 6 bytes.

Q11. What is pipelining?

Pipelining is a technique in which instruction fetching and execution overlap to improve processor efficiency.

Q12. What is the memory capacity of 8086?

8086 supports 1 MB of memory because it has a 20-bit address bus.

Numerical Questions to Practice

Week 6 is mainly conceptual, but you should prepare these architecture-based numerical questions.

Q1. Calculate the addressable memory capacity of 8086.

Given:

- Address bus = 20 bits
- Addressable unit = 1 byte

Formula:

$$\text{Memory Capacity}=2^n$$
$$=2^{20}=1,048,576\text{ bytes}$$

Answer:

$$\boxed{1\text{ MB}}$$

Q2. How many different bit patterns can a 16-bit register store?

$$2^{16}=65,536$$

Answer: 65,536 different bit patterns.

Q3. If a processor has a 24-bit address bus, how many memory locations can it address?

$$2^{24}=16,777,216$$

If each location stores one byte:

$$\boxed{16\text{ MB}}$$

Week 6 Practice MCQs

Test yourself

0/12 answered

1. Which type of execution follows the normal order of instructions?

Branching
Sequential execution
Conditional execution
Microprogramming

2. What happens during an unconditional jump?

A condition is checked
The PC changes to the target address
The ALU is disabled
The instruction queue becomes larger

3. Which control unit uses fixed logic circuits?

Microprogrammed
Hardwired
Software-controlled
Memory-controlled

4. Which control unit uses microinstructions?

Hardwired
Microprogrammed
ALU
BIU

5. 8086 is a _____ microprocessor.

8-bit
16-bit
20-bit
32-bit

6. What is the address bus width of 8086?

8-bit
16-bit

20-bit
32-bit

7. What is the maximum memory capacity of 8086?

64 KB
256 KB
512 KB
1 MB

8. Which unit fetches instructions from memory?

EU
ALU
BIU
Flag Register

9. What is the size of the 8086 instruction queue?

2 bytes
4 bytes
6 bytes
8 bytes

10. Which unit decodes and executes instructions?

BIU
EU
Memory Unit
Address Bus

11. What is the main benefit of pipelining?

Increases memory size
Improves instruction processing efficiency
Reduces address bus width
Removes the ALU

12. What happens to the 8086 instruction queue after a taken branch?

It doubles in size
It is flushed and refilled from the new address
It becomes permanent
It stores the result
Submit answers

Final Revision Checklist

Before moving to Week 7, make sure you can answer these without looking at the notes:

- [ ] Difference between sequential execution and branching.
- [ ] Conditional and unconditional branching.
- [ ] Functions of control signals.
- [ ] Hardwired vs. microprogrammed control.
- [ ] Introduction and features of 8086.
- [ ] Why 8086 is called a 16-bit processor.
- [ ] Why 8086 supports 1 MB memory.
- [ ] Functions of BIU and EU.
- [ ] Difference between BIU and EU.
- [ ] Six-byte instruction queue.
- [ ] Pipelining and its benefits.
- [ ] Complete 8086 block diagram.

Most important diagrams to practice: Hardwired Control Unit, Microprogrammed Control Unit, and the complete 8086 architecture showing BIU, EU, and the instruction queue.